EXPERIMENT NO. 4

Name: Ayush Rahul Kulkarni

Roll no: B26ET2024 Div: SY3 Class: C3

Input Code:

```
#include <stdio.h>
#include <string.h>
#define MAX 100
struct Employee {
    int id;
    char name[50];
    char designation[50];
    float salary;
};
void createDatabase(struct Employee emp[], int *n) {
    printf("Enter the number of employees to create: ");
    scanf("%d", n);
    for (int i = 0; i < *n; i++) {
        printf("\nEnter details for employee %d:\n", i + 1);
        printf("ID: ");
        scanf("%d", &emp[i].id);
        printf("Name: ");
        scanf(" %[^\n]s", emp[i].name);
        printf("Designation: ");
        scanf(" %[^\n]s", emp[i].designation);
        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }
    printf("\nDatabase created successfully!\n");
}
void displayDatabase(struct Employee emp[], int n) {
```

```
    if (n == 0) {
        printf("\nDatabase is empty!\n");
        return;
    }
    printf("\n--- Employee Database ---\n");
    printf("%-5s | %-20s | %-20s | %-10s\n", "ID", "Name", "Designation", "Salary");
    printf("-----------------------------------------------------------\n");
    for (int i = 0; i < n; i++) {
        printf("%-5d | %-20s | %-20s | %-10.2f\n", emp[i].id, emp[i].name, emp[i].designation,
emp[i].salary);
    }
}
void modifyEmployee(struct Employee emp[], int n) {
    int id, found = 0;
    printf("\nEnter Employee ID to modify: ");
    scanf("%d", &id);
    for (int i = 0; i < n; i++) {
        if (emp[i].id == id) {
            printf("Enter new Name: ");
            scanf(" %[^\n]s", emp[i].name);
            printf("Enter new Designation: ");
            scanf(" %[^\n]s", emp[i].designation);
            printf("Enter new Salary: ");
            scanf("%f", &emp[i].salary);
            printf("Employee details updated successfully!\n");
            found = 1;
            break;
        }
    }
    if (!found) {
        printf("Employee with ID %d not found.\n", id);
```

```
    }
}
void appendEmployee(struct Employee emp[], int *n) {
    if (*n >= MAX) {
        printf("\nDatabase is full! Cannot append more employees.\n");
        return;
    }
    printf("\nEnter details for new employee:\n");
    printf("ID: ");
    scanf("%d", &emp[*n].id);
    printf("Name: ");
    scanf(" %[^\n]s", emp[*n].name);
    printf("Designation: ");
    scanf(" %[^\n]s", emp[*n].designation);
    printf("Salary: ");
    scanf("%f", &emp[*n].salary);
    (*n)++;
    printf("Employee appended successfully!\n");
}
void searchEmployee(struct Employee emp[], int n) {
    int id, found = 0;
    printf("\nEnter Employee ID to search: ");
    scanf("%d", &id);
    for (int i = 0; i < n; i++) {
        if (emp[i].id == id) {
            printf("\nEmployee Found:\n");
            printf("ID: %d\nName: %s\nDesignation: %s\nSalary: %.2f\n", emp[i].id,
emp[i].name, emp[i].designation, emp[i].salary);
            found = 1;
            break;
        }
```

```
    }
    if (!found) {
        printf("Employee with ID %d not found.\n", id);
    }
}
void sortEmployees(struct Employee emp[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (emp[j].id > emp[j + 1].id) {
                struct Employee temp = emp[j];
                emp[j] = emp[j + 1];
                emp[j + 1] = temp;
            }
        }
    }
    printf("\nEmployees sorted successfully by ID!\n");
}
int main() {
    struct Employee emp[MAX];
    int n = 0, choice;
    do {
        printf("\n--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---\n");
        printf("1. Create Database\n");
        printf("2. Display Database\n");
        printf("3. Modify Employee\n");
        printf("4. Append Employee\n");
        printf("5. Search Employee\n");
        printf("6. Sort Employees\n");
        printf("7. Exit\n");
        printf("Enter your choice: ");
```

```
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                createDatabase(emp, &n);
                break;
            case 2:
                displayDatabase(emp, n);
                break;
            case 3:
                modifyEmployee(emp, n);
                break;
            case 4:
                appendEmployee(emp, &n);
                break;
            case 5:
                searchEmployee(emp, n);
                break;
            case 6:
                sortEmployees(emp, n);
                break;
            case 7:
                printf("Exiting program. Goodbye!\n");
                break;
            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 7);
    return 0;
}
```

Output:

```
--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 1
Enter the number of employees to create: 2

Enter details for employee 1:
ID: 235064
Name: karan
Designation: jr game developer
Salary: 600000

Enter details for employee 2:
ID: 235065
Name: arjun
Designation: web developer
Salary: 1200000

Database created successfully!

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 2

--- Employee Database ---
ID | Name | Designation | Salary
-----------------------------------------------------------------
235064 | karan | jr game developer | 600000.000000
235065 | arjun | web developer | 1200000.000000

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
```

```
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 3

Enter Employee ID to modify: 235064
Enter new Name: karan
Enter new Designation: sr game developer
Enter new Salary: 1200000
Employee details updated successfully!

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 2

--- Employee Database ---
ID | Name | Designation | Salary
-----------------------------------------------------------------
235064 | karan | sr game developer | 1200000.000000
235065 | arjun | web developer | 1200000.000000

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 4

Enter details for new employee:
ID: 235066
Name: salman
Designation: jr game developer
Salary: 600000
Employee appended successfully!

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
```

```
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 2

--- Employee Database ---
ID | Name | Designation | Salary
-----------------------------------------------------------------
235064 | karan | sr game developer | 1200000.000000
235065 | arjun | web developer | 1200000.000000
235066 | salman | jr game developer | 600000.000000

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 5

Enter Employee ID to search: 235065

Employee Found:
ID: 235065
Name: arjun
Designation: web developer
Salary: 1200000.00

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 3

Enter Employee ID to modify: 235066
Enter new Name: hamza
Enter new Designation: jr game developer
```

```
Enter new Designation: jr game developer
Enter new Salary: 600000
Employee details updated successfully!

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 6

Employees sorted successfully by ID!

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 2

--- Employee Database ---
ID | Name | Designation | Salary
-----------------------------------------------------------------
235064 | karan | sr game developer | 1200000.000000
235065 | arjun | web developer | 1200000.000000
235066 | hamza | jr game developer | 600000.000000

--- EMPLOYEE DATABASE MANAGEMENT SYSTEM ---
1. Create Database
2. Display Database
3. Modify Employee
4. Append Employee
5. Search Employee
6. Sort Employees
7. Exit
Enter your choice: 7
Exiting program. Goodbye!
```